

## Application interfaces

An IoT solution consists of some Web application interfaces like the Admin Information Dashboard or Cockpit. This Cockpit provides the interface to administer the IoT devices, as well as the configurations to manage and control the IoT devices. There can be multiple Web application interfaces. One can be at each deployment unit, which represents the edge computation at the periphery of the system. There can be other interfaces that manage the entire system's configurations from a central portal.

The Cockpit can be implemented using front-end technologies like Angular, Node, etc.

- **Angular:** This is a JavaScript based open source, front-end Web application framework. It supports the model-view-controller (MVC) and the model-view-view-model (MVVM) architectures. It aims to simplify the development and testing of single page Web applications.
- **Node:** This is an open source, cross-platform, JavaScript runtime environment for executing JavaScript code on the server side. It has an event-driven architecture and its design choices aim to optimise throughput and scalability in the Web applications.

## Back-end platform and the open source technologies used

An IoT solution's back-end system has the following functions.

- **Ingestion:** An ingestion framework is needed in the solutions approach to extract data from various data sources and send it to the processing tools. Data can be ingested as a stream in real-time or ingested on batch processing, depending on the business' needs. An ingestion of data to the system for further processing requires various supportive frameworks based on other technology stacks of the system. The data input to the system can be from Hadoop data clusters, SQL data exports or data ingestion to the messaging server like Kafka, or other data stream processing frameworks/tools like Apache Camel, Spark Streaming, Storm and Flume.
- **Orchestration and processing:** This is the layer in which the alignment of business needs to applications, data and infrastructure takes place. System orchestration and automation are important layers of the entire

solution. They define various rules, policies, business logic, automated workflows, provisioning, change management, etc. The entire system's scalability depends a lot on this layer's scalability and extensibility provisioning capabilities. There are many frameworks or tools that can be used for system orchestration and processing in enterprises with large data, like NiFi, Oozie and Apache Spark.

- **Modelling and analytics:** Data modelling for organisational data can span multiple levels of abstraction, ranging from conceptual to logical and physical. The conceptual models can be used for discussions with business people and domain experts. The logical model adds more precision, and provides the information to discuss and decide on logical representations. The physical model relies on various technology-specific data, and helps to prepare a target environment, such as a database management system. There are various frameworks or technologies available for data modelling and the respective analytics, like R, Python, Spark ML, Kibana, Elastic Search and TensorFlow.
- **Data storage:** Data storage components are the core of any solutions approach. From raw data to mission-critical records, the choice of storage can have a profound impact on the capacity, performance, long-term reliability and durability of any storage infrastructure. The system should have failover, backup and disaster recovery mechanisms. Data generated from various data sources may be structured, unstructured or semi-structured. Solutions should have a provision to process, format and message (if needed), and then provide the relevant storage options like SQL, NOSQL, a data warehouse, etc. The data storage platform depends on the nature of the data and the business needs. Some of the open source data storage options are MongoDB, MySQL, PostgreSQL, Redis, Couch DB, HBase, and Cassandra.

## Reference architecture of edge computing for IoT solutions with the cloud technology stack

Cloud service providers have rich sets of services and technology options for IoT applications and edge computing. Considering the leading cloud service providers as Azure and AWS, Figure 3 depicts the technology choices at various layers of reference architecture.

The following are the cloud technologies and services available.

- **Azure IoT Hub:** This is a fully managed service that enables reliable and secure bi-directional communication between millions of IoT devices and a solutions back-end. It provides multiple device-to-cloud and cloud-to-device communication options. These options include one-way messaging, file transfers and request-reply methods. It has built-in

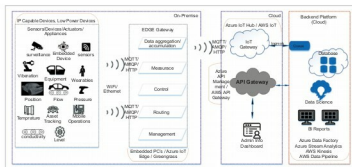


Figure 3: Reference architecture for cloud services